

● RESEARCH GRADE · 99%+ PURITY · USA-BASED

RESEARCH REFERENCE & SAFETY DOCUMENTATION

Semax

Synthetic ACTH(4–10) analogue heptapeptide · Research-grade reference material

SEQUENCE

MEHFPGP

FORMULA

$C_{37}H_{51}N_9O_{10}S$

MW

813.9 g/mol

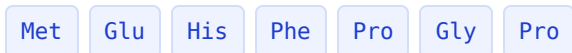
CLASS

Neuropeptide (7 aa)

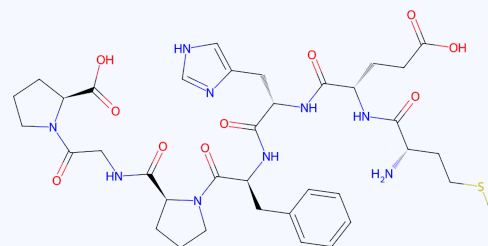
Identity & Composition

Semax is a synthetic heptapeptide comprising the ACTH(4–7) fragment extended with a C-terminal Pro-Gly-Pro tripeptide, which increases stability. Developed at the Institute of Molecular Genetics, Moscow. It lacks the hormonal activity of ACTH. Supplied lyophilized, research grade.

Product name	Semax
Class	Synthetic ACTH(4–10) analogue
Sequence	Met-Glu-His-Phe-Pro-Gly-Pro
Length	7 amino acids
Molecular formula	C ₃₇ H ₅₁ N ₉ O ₁₀ S
Molecular weight	813.9 g/mol
Structural note	ACTH(4–7) + C-terminal Pro-Gly-Pro
Appearance	White lyophilized powder
Identified use	In-vitro research & development only



Primary sequence (N → C): MEHFPGP



Molecular structure. 2D structure computed from the primary sequence; formula and mass consistent with C₃₇H₅₁N₉O₁₀S (813.9 g/mol). The C-terminal Pro-Gly-Pro extension confers resistance to enzymatic degradation relative to the parent ACTH fragment.

Note on classification. Although derived from an ACTH fragment, Semax is reported in the literature to lack ACTH hormonal (corticotropic) activity — it is studied as a neuroactive peptide rather than a hormone analogue in the endocrine sense.

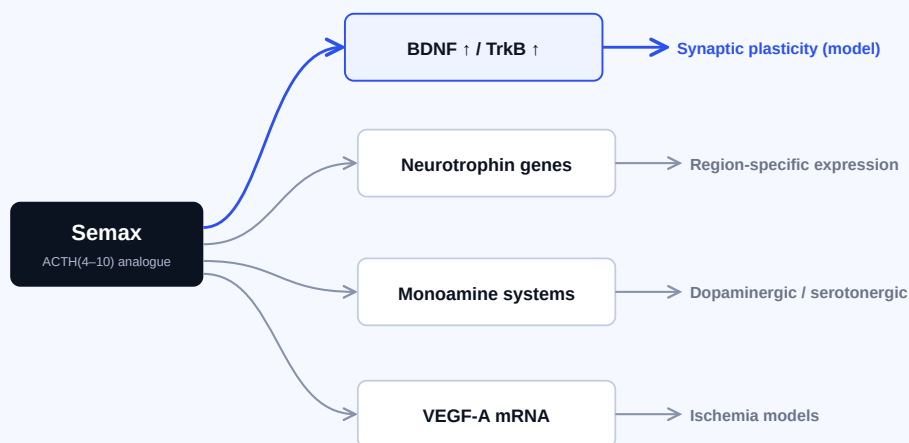
Research Background

Semax has been investigated since the 1980s–1990s, predominantly by Russian research groups, in neuroprotection, cerebral ischemia and cognition models. The C-terminal Pro-Gly-Pro extension distinguishes it from the parent ACTH fragment by conferring greater metabolic stability.

Mechanisms studied in preclinical models

Findings below derive from animal and cell-culture systems. They are not established effects in humans.

- **BDNF / TrkB modulation.** Dolotov et al. (2006) reported increased BDNF protein and TrkB phosphorylation, with elevated exon III BDNF and trkB mRNA, in rat hippocampus.
- **Neurotrophin gene expression.** Rapid, gene- and region-specific changes in neurotrophin expression reported in rat brain.
- **Ischemia models.** Activation of neurotrophic factor expression and modulation of VEGF-A mRNA in frontal cortex and hippocampus in cerebral ischemia models; reduced infarct volume reported.
- **Monoaminergic modulation.** Reported effects on dopaminergic, serotonergic and cholinergic systems.
- **Melanocortin receptor interaction.** Described in the literature as an MC4 receptor antagonist while lacking ACTH corticotropic activity.



Proposed pathways investigated in preclinical (animal / in-vitro) models. Schematic only; not a claim of established human effect.

What the Literature Supports — and What It Doesn't

Semax occupies an unusual position: it is an approved medicine in Russia, yet has not been approved by the FDA or EMA, and large-scale randomized controlled trial data available in Western indexed literature remains limited. Much of the primary research is Russian-published.

Evidence at a glance

PRECLINICAL

Substantial

HUMAN RCT (WESTERN)

Limited

RUSSIA

Approved medicine

US FDA / EMA

Not approved

- The BDNF/TrkB mechanism is supported by **specific, quantified preclinical findings** (rat hippocampus), not merely asserted.
- A **large share of the literature originates from a single national research tradition**, with less independent Western replication than the volume of publications alone might suggest.
- **Regulatory divergence is itself informative:** approval in one jurisdiction alongside non-approval in the US and EU indicates the evidence base has not met FDA/EMA standards.
- Reported human use in stroke and cognitive-disorder contexts should not be read as establishing efficacy under Western regulatory evidentiary standards.

Appropriate characterization: a neuroactive peptide with a well-described preclinical neurotrophic mechanism and a national regulatory approval, but without the independent large-scale controlled human evidence required for FDA or EMA authorization.

Regulatory & status notes

US FDA	Not approved for any use.
Russia	Reported as approved for stroke and cognitive indications. Approval in one jurisdiction does not imply FDA/EMA acceptance.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

1. Dolotov OV, et al. Semax, an analog of ACTH(4-10) with cognitive effects, regulates BDNF and trkB expression in the rat hippocampus. Brain Res. 2006;1117(1):54–60. PMID 16996037.
2. Dolotov OV, et al. Semax, an analogue of adrenocorticotropin (4-10), binds specifically and increases levels of BDNF protein in rat basal forebrain. J Neurochem. 2006;97 Suppl 1:82–86.
3. Adan RA, et al. Identification of antagonists for melanocortin MC3, MC4 and MC5 receptors. Eur J Pharmacol. 1994;269:331–7.
4. Semax peptide targets Oprm1 to promote deubiquitination and functional recovery after spinal cord injury (murine model).

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	Semax
Sequence	MET - GLU - HIS - PHE - PRO - GLY - PRO
Formula / MW	C37H51N9O10S / 813.9
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use
Supplier	
Emergency	

2 Hazard Identification

Not classified based on currently available data. Limited toxicological data; absence of classification is not a determination of absence of hazard. **Signal word:** None. **Pictograms:** None required. **Precautionary:** P261, P264, P280, P501. Handle as a potentially bioactive peptide of unknown toxicity.

3 Composition

Single-substance product, >98% research grade. Residual synthesis reagents, solvents and counter-ions may be present consistent with research-grade material. See lot COA for impurity profile.

4 First-Aid Measures

Eyes: rinse with water. **Skin:** wash with soap and water; remove contaminated clothing. **Inhalation:** fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Treat symptomatically; no specific antidote known.

5 Fire-Fighting

Media: CO₂, dry chemical, foam, water spray. Combustion may produce CO, CO₂, NO_x and sulfur oxides (methionine-containing). Use SCBA and full protective gear.

6 Accidental Release

Avoid dust; ventilate; use PPE per §8. Prevent entry to drains/waterways. Collect in a closed labeled container for disposal per §13.

7 Handling & Storage

Handle as bioactive peptide of unknown toxicity; minimize exposure. Weigh/reconstitute in fume hood. Store lyophilized solid tightly closed, protected from light/moisture/heat; -20 °C long-term, 2-8 °C short-term. Aliquot reconstituted solutions; store frozen, protected from light. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Fume hood for dry powder. Nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator where controls insufficient. Eyewash/safety shower per ANSI Z358.1.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	White powder
Odor	Odorless
Solubility	Soluble in water
MW	813.9 g/mol
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable under normal storage. Avoid heat, moisture, sunlight, UV, open flame. Incompatible with strong oxidizers, strong acids/bases, strong reducing agents. The methionine residue is susceptible to oxidation. Decomposition: CO, CO₂, NO_x, SO_x. No hazardous polymerization.

11 Toxicological Information

Not fully characterized. No LD₅₀/LC₅₀ available from authoritative databases. Skin/eye: treat as potentially irritating. Sensitization: cannot be ruled out. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: no authoritative substance-specific data; not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

No substance-specific ecotoxicity, persistence or bioaccumulation data identified. Avoid release to surface water, soil, drains or sewers.

13 Disposal

Dispose per applicable local/state/federal regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not regulated as dangerous goods under DOT (49 CFR), IATA or IMDG based on current classification. UN number: not applicable. Shipper must independently verify.

15 Regulatory Information

US TSCA: may qualify for R&D exemption (40 CFR 720.36). OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption. Note: regulatory status differs by jurisdiction (see Section 03 of the research reference). Not a legal determination of regulatory status.

16 Other Information

Revision [DATE] · Version 1.0. Independent professional review recommended prior to use. Sources: PubChem; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8; OECD Test Guidelines. Provided without warranty. For scientific R&D use only.

Disclaimer. Provided for laboratory research reference only. Summarizes published preclinical literature and standardized safety information; does not constitute medical, therapeutic, or usage guidance, nor describe or recommend administration to humans or animals. Compiled from cited sources believed accurate as of the revision date, without warranty. Users bear sole responsibility for verifying all information, ensuring safe handling, and complying with applicable regulations. Sold for laboratory research purposes only. Not for human or veterinary use. Must be 21+ to purchase.